Medical Cost Analysis & Linear Regression Using Python

```
A medical expenses are any money that is spent for the
prevention or treatment of injury, sickness or desease. these
expenses include health and dental insurance premiums, doctor
and hospital visit, prescription and over the counter drugs,
glasses and contacts, crutches and wheelchairs and other
medical related expenses that are not included. health care
cost have been a major concern for both patients and
insurance companies. with increase in medical expenses,
understanding the key factor that drive these cost is
essential for developing more efficient healthcare plans and
managing financial risk.
```

OBJECTIVE

```
this project aims at analyzing the factors that drive health
care expenses in United State of America
```

Data Source: Kaggle

let's import our python libraries

In [1]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error
import warnings
warnings.filterwarnings('ignore')
```

loading our data set

In [2]:
```
path = r"C:\Users\USER\Desktop\csv\health.csv"
```

In [3]:
```
df = pd.read_csv(path)
df
```

Out[3]:

	s/n	age	sex	bmi	children	smoker	region	medical charges
0	1	19	Female	27.900	0	Yes	Southwest	16884.92400
1	2	18	Male	33.770	1	No	Southeast	1725.55230
2	3	28	Male	33.000	3	No	Southeast	4449.46200
3	4	33	Male	22.705	0	No	Northwest	21984.47061
4	5	32	Male	28.880	0	No	Northwest	3866.85520
...	...	...	...	...	...	...	...	...
1333	1334	50	Male	30.970	3	No	Northwest	10600.54830
1334	1335	18	Female	31.920	0	No	Northeast	2205.98080
1335	1336	18	Female	36.850	0	No	Southeast	1629.83350
1336	1337	21	Female	25.800	0	No	Southwest	2007.94500
1337	1338	61	Female	29.070	0	Yes	Northwest	29141.36030

1338 rows × 8 columns

let's understand our dataset

AGE = Age of an individual/primary beneficiary

SEX = Gender of an individual

BMI = Body Mass Index of an individual

CHILDREN = Number of children of an individual covered by the insurance

SMOKER = Whether the individual smokes or not (yes/no)

REGION = The individual's residential area in the US, northeast, southeast, southwest, northwest.

MEDICAL CHARGE = Medical cost billed by the insurance

pre data processing: data wrangling/cleaning

check our data type to see whether its correct

In [4]:

```
df.dtypes
```

Out[4]:
```
s/n                  int64
age                  int64
sex                 object
bmi                float64
children             int64
smoker              object
region              object
medical charges    float64
dtype: object
```

identifying and handling missing value

In [5]:
```
df.replace("?", np.nan, inplace = True)
```

detecting a missing value

In [6]:
```
df.isnull()
```

Out[6]:

	s/n	age	sex	bmi	children	smoker	region	medical charges
0	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...
1333	False	False	False	False	False	False	False	False
1334	False	False	False	False	False	False	False	False
1335	False	False	False	False	False	False	False	False
1336	False	False	False	False	False	False	False	False
1337	False	False	False	False	False	False	False	False

1338 rows × 8 columns

In [7]:
```
df.isnull().sum()
```

Out[7]:
```
s/n                0
age                0
sex                0
bmi                0
children           0
smoker             0
region             0
medical charges    0
dtype: int64
```

there is no any missing value.

our data is ready for use

descriptive statistics analysis

In [8]:
```
df.describe()
```

Out[8]:

	s/n	age	bmi	children	medical charges
count	1338.000000	1338.000000	1338.000000	1338.000000	1338.000000
mean	669.500000	39.207025	30.663397	1.094918	13270.422265
std	386.391641	14.049960	6.098187	1.205493	12110.011237
min	1.000000	18.000000	15.960000	0.000000	1121.873900
25%	335.250000	27.000000	26.296250	0.000000	4740.287150
50%	669.500000	39.000000	30.400000	1.000000	9382.033000
75%	1003.750000	51.000000	34.693750	2.000000	16639.912515
max	1338.000000	64.000000	53.130000	5.000000	63770.428010

In [9]:
```
df.describe(include = 'all')
```

Out[9]:

	s/n	age	sex	bmi	children	smoker	region
count	1338.000000	1338.000000	1338	1338.000000	1338.000000	1338	1338
unique	NaN	NaN	2	NaN	NaN	2	4
top	NaN	NaN	Male	NaN	NaN	No	Southeast
freq	NaN	NaN	676	NaN	NaN	1064	364
mean	669.500000	39.207025	NaN	30.663397	1.094918	NaN	NaN
std	386.391641	14.049960	NaN	6.098187	1.205493	NaN	NaN
min	1.000000	18.000000	NaN	15.960000	0.000000	NaN	NaN
25%	335.250000	27.000000	NaN	26.296250	0.000000	NaN	NaN
50%	669.500000	39.000000	NaN	30.400000	1.000000	NaN	NaN
75%	1003.750000	51.000000	NaN	34.693750	2.000000	NaN	NaN
max	1338.000000	64.000000	NaN	53.130000	5.000000	NaN	NaN

- Here we can see that there are 1338 values in each of our variable and this indicate that there is no missing values

LET'S LOOK AT THE MEAN(AVERAGE) VALUE AND THE STANDARD DEVIATION

- The average age of individual is approximately 39 years and the standard deviation is aproximately 14. this mean, on average, the age of an individual deviate from the average age with 14 year. which implies that some individual might be about 14 years younger than the average age while others might be 14 years older than the average age

- The average individual BMI is 30 and the standard deviation is aproximately 6. this implies that, individual BMI deviate from that average with approximately 6 BMI. this is to say some individual might have around 6 BMI lower from the average BMI, while some may have around 6 BMI higher from the average.
- On average the number of children is aproximately 1 and the standard deviation is approximately 1. this implies that the number of children deviate by one from the average number, in other word, some individual might have about 1 child less than average while others may have 1 child more than the average
- The average medical charge is aproximately 13270, the standard deviation is approximately 12110, this implies that the medical charges deviates from the mean with approximately 12110. in other word individual might be charge around 12110 lower from the average charge while others might be charge about 12110 higher from the average
- THE MAXIMUM AND THE MINIMUM
- The highest age is 64 and the lowest age is 18
- The highest number of children which an individual have is 5 and the lowest is 0

FREQUENCY

- Male gender have appear more with 676 time and also the southeast region is the region that appear most with 364 time

LET'S LOOK AT THE QUARTILES and DATA DISTRIBUTION

FOR AGE

- 25% (Q1) 25% of the individual are below 27 years
- 50% (Q1) 50% of individual are below 39 years
- 75% (Q1) 75% of individuals are below 51 years

FOR BMI

- 25% (Q1) 25% of individuals has aproximately 26 BMI
- 50% (Q1) 50% of individuals has aproximately 30 BMI
- 75% (Q1) 75% of individuals have aproximately 53 BMI

FOR CHILDREN

- 25% (Q1) 25% of individuals have 0 children(they do not have a child)
- 50% (Q1) 50% of individual have 1 child
- 75% (Q1) 75% of individaul have 2 child

FOR MEDICAL CHARGE

- 25% (Q1) 25% of individual are been charge aproximately 4740
- 50% (Q1) 50% of individual are charge aproximately 9382
- 75% (Q1) 75% of individual are charge aproximately 16639

LET'S TRANSFORM OUR VARIABLE VALUE(SMOKER) INTO NUMERICAL VARIABLE

In [10]:
```
data = {'smoker': ['Yes', 'No', 'No', 'Yes', 'Yes']}
```

In [11]:
```
df['smoker_encoded'] = df['smoker'].map({'Yes': 1, 'No': 0})
df
```

Out[11]:

	s/n	age	sex	bmi	children	smoker	region	medical charges	smoker_enc
0	1	19	Female	27.900	0	Yes	Southwest	16884.92400	
1	2	18	Male	33.770	1	No	Southeast	1725.55230	
2	3	28	Male	33.000	3	No	Southeast	4449.46200	
3	4	33	Male	22.705	0	No	Northwest	21984.47061	
4	5	32	Male	28.880	0	No	Northwest	3866.85520	
...	...	...	...	...	...	...	...	...	
1333	1334	50	Male	30.970	3	No	Northwest	10600.54830	
1334	1335	18	Female	31.920	0	No	Northeast	2205.98080	
1335	1336	18	Female	36.850	0	No	Southeast	1629.83350	
1336	1337	21	Female	25.800	0	No	Southwest	2007.94500	
1337	1338	61	Female	29.070	0	Yes	Northwest	29141.36030	

1338 rows × 9 columns

LET'S REARRANGE OUR COLUMN

i want smoker encoded to come after smoker

In [12]:
```
data = {'age': [1, 2, 3], 'sex': [4, 5, 6], 'bmi': [7, 8, 9], 'children': [10, 11,
```

In [13]:
```
df = df[['age', 'sex', 'bmi', 'children', 'smoker', 'smoker_encoded', 'region', 'r
```

In [14]:
```
df
```

Out[14]:

	age	sex	bmi	children	smoker	smoker_encoded	region	medical charges
0	19	Female	27.900	0	Yes	1	Southwest	16884.92400
1	18	Male	33.770	1	No	0	Southeast	1725.55230
2	28	Male	33.000	3	No	0	Southeast	4449.46200
3	33	Male	22.705	0	No	0	Northwest	21984.47061
4	32	Male	28.880	0	No	0	Northwest	3866.85520
...	...	...	...	...	...	...	...	...
1333	50	Male	30.970	3	No	0	Northwest	10600.54830
1334	18	Female	31.920	0	No	0	Northeast	2205.98080
1335	18	Female	36.850	0	No	0	Southeast	1629.83350
1336	21	Female	25.800	0	No	0	Southwest	2007.94500
1337	61	Female	29.070	0	Yes	1	Northwest	29141.36030

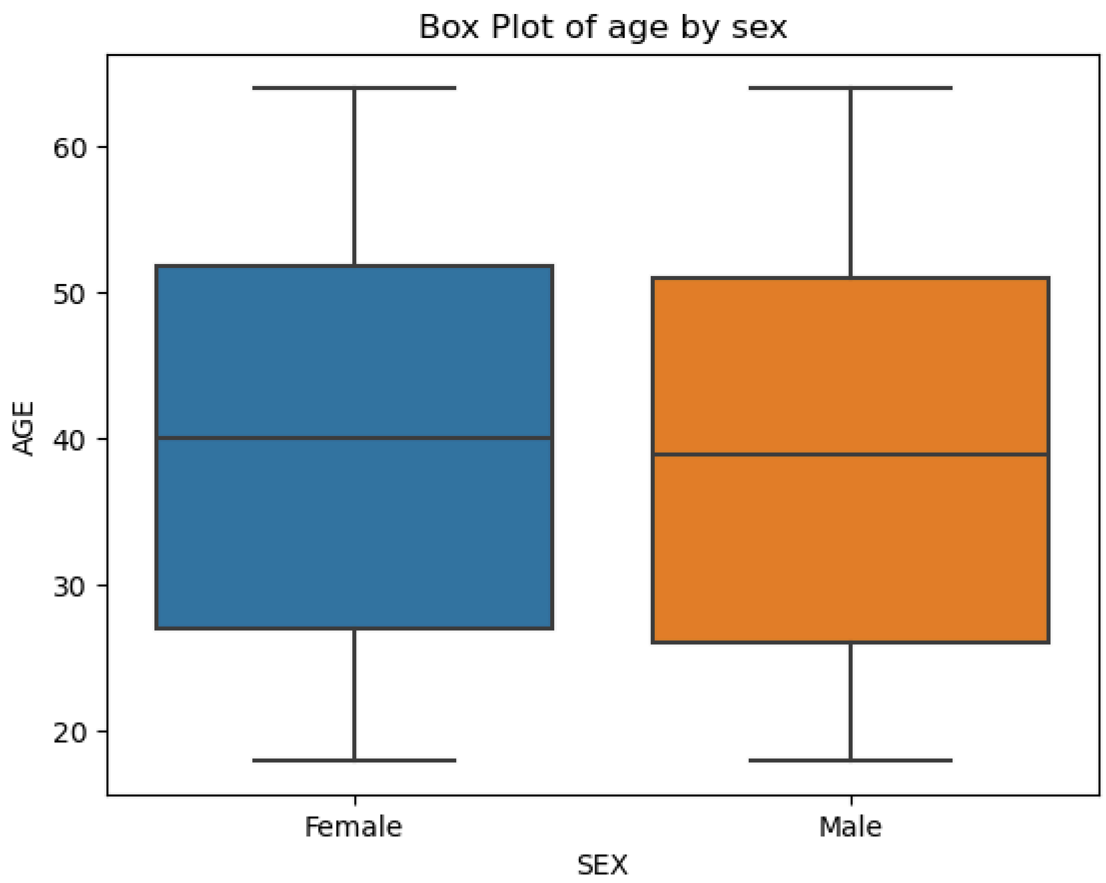
1338 rows × 8 columns

In [15]:

```
sns.boxplot(x=df['sex'], y=df['age'])

plt.title('Box Plot of age by sex')
plt.xlabel('SEX')
plt.ylabel('AGE')

plt.show()
```

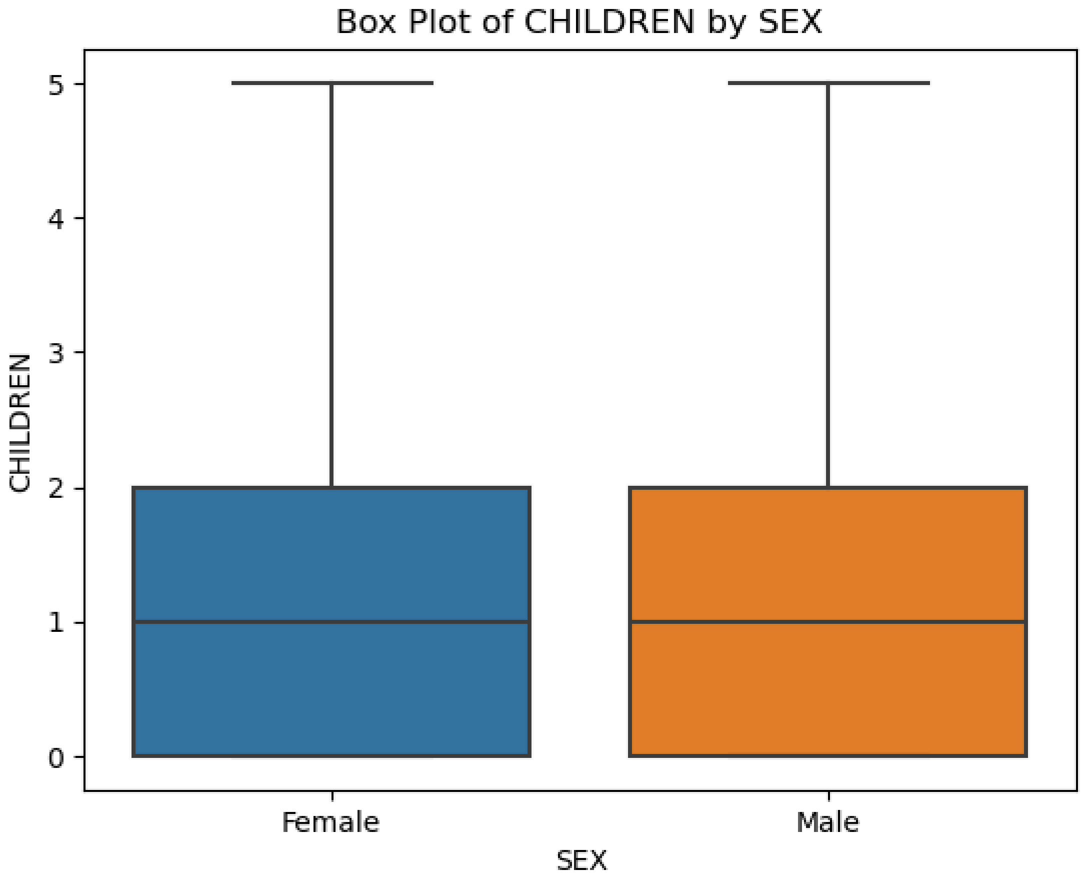


- we can see where our minimum data range lies(upper whisker line)
- we can see where our (Q1) (25th percentile) which is the bottom line of the box
- we can see where our (Q2) (50th percentile) which is the line inside the box
- we can see where our (Q3) (25th percentile) which is the top line of the box
- we can see where our maximum range data lies(lower whisker line)
- this data is free of outliers

In [16]:
```
sns.boxplot(x=df['sex'], y=df['children'])

plt.title('Box Plot of CHILDREN by SEX')
plt.xlabel('SEX')
plt.ylabel('CHILDREN')

plt.show()
```

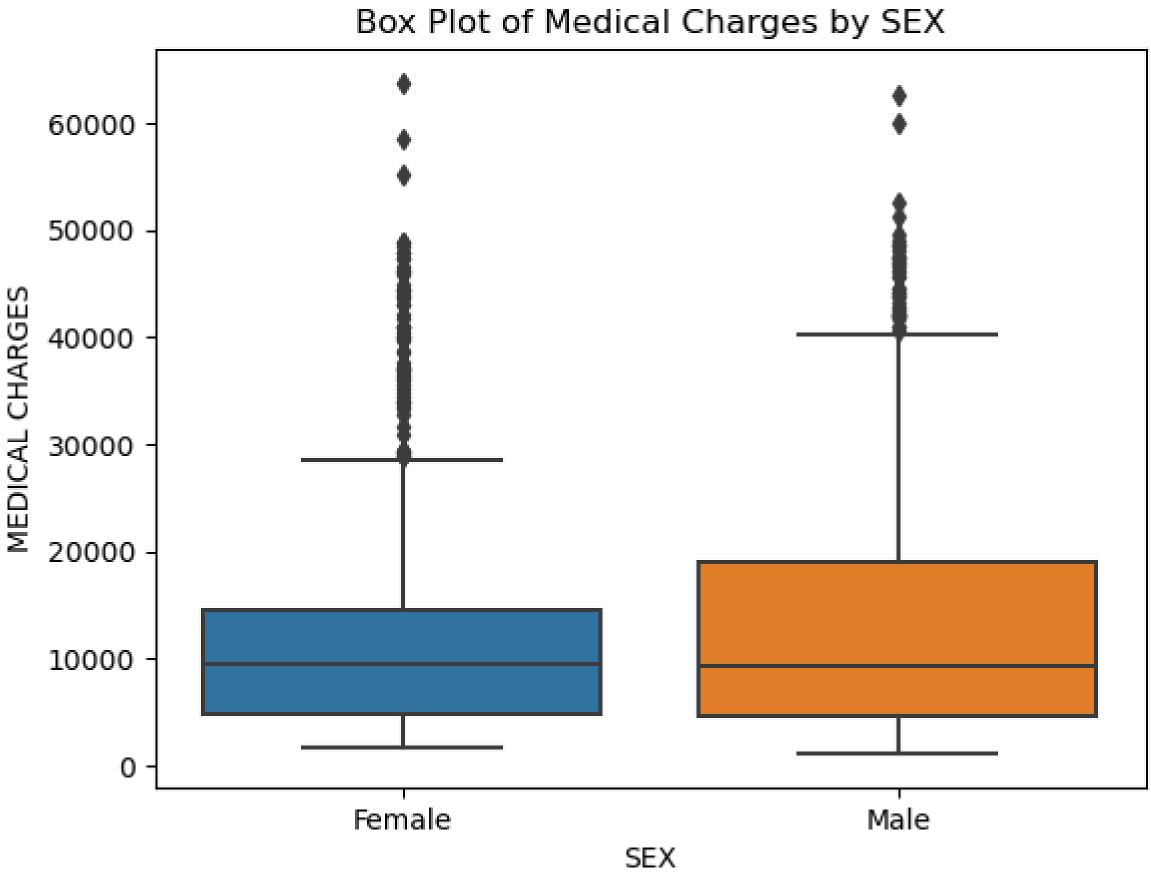


- we can see where our (Q1) (25th percentile) which is the bottom line of the box
- we can see where our (Q2) (50th percentile) which is the line inside the box
- we can see where our (Q3) (25th percentile) which is the top line of the box
- we can see where our maximum range data lies(upper whisker line)
- this data is free of outliers

In [17]:
```
sns.boxplot(x=df['sex'], y=df['medical charges'])

plt.title('Box Plot of Medical Charges by SEX')
plt.xlabel('SEX')
plt.ylabel('MEDICAL CHARGES')

plt.show()
```

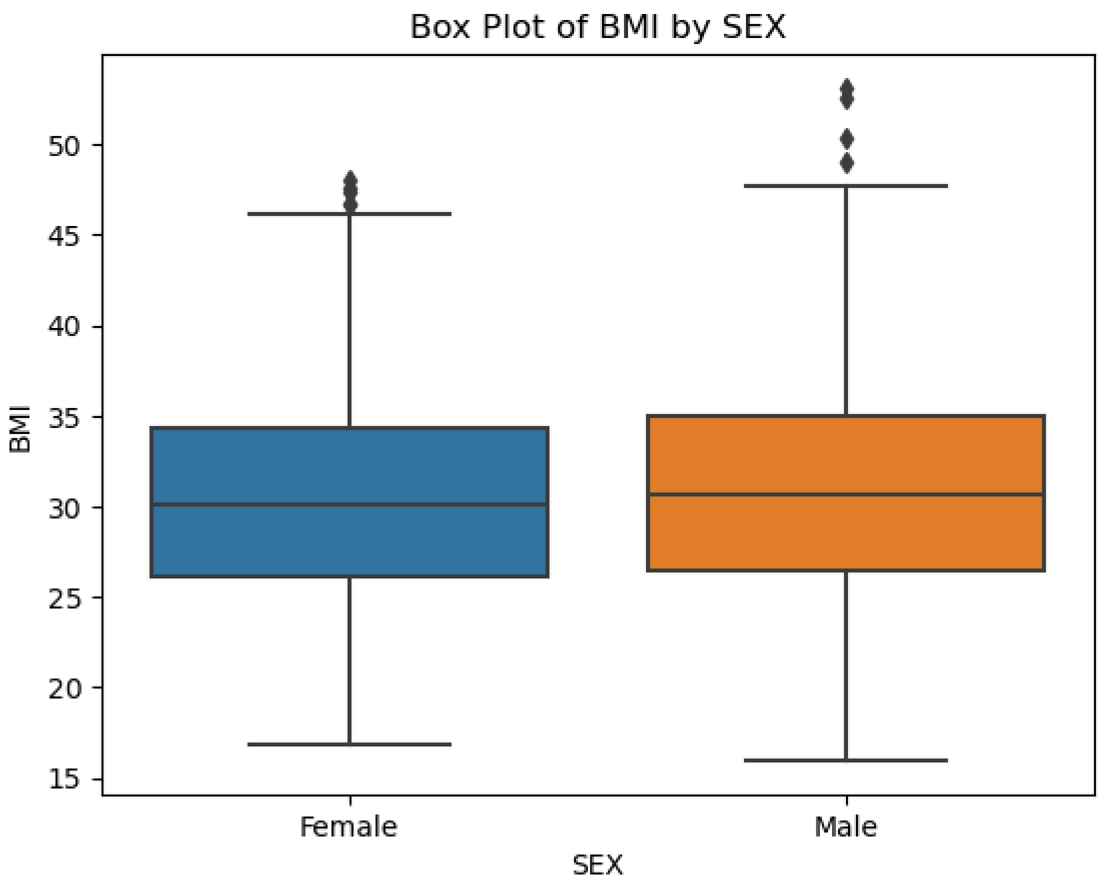


- we can see where our minimum range data lies
- we can see where our (Q1) (25th percentile) which is the bottom line of the box
- we can see where our (Q2) (50th percentile) which is the line inside the box
- we can see where our (Q3) (25th percentile) which is the top line of the box
- we can see where our maximum range data lies
- this data have a lot of outliers, and this could indicate there are individuals with exceptionally high healthcare costs, possibly due to specific health conditions

In [18]:
```
sns.boxplot(x=df['sex'], y=df['bmi'])

plt.title('Box Plot of BMI by SEX')
plt.xlabel('SEX')
plt.ylabel('BMI')

plt.show()
```



- we can see where our minimum range data lies(lower whisker line)
- we can see where our (Q1) (25th percentile) which is the bottom line of the box
- we can see where our (Q2) (50th percentile) which is the line inside the box
- we can see where our (Q3) (25th percentile) which is the top line of the box
- we can see where our maximum data range lies(upper whisker line)
- this data have outliers we can see it, Points above the upper Whiskers line. this could indicate individuals with exceptionally high bmi

some of our data have a lot of outlier especially the medical charges dataset, and it is obvious that all the observation in that variables are important. Therefore we will use winsorizing method to manage our medical charges data set

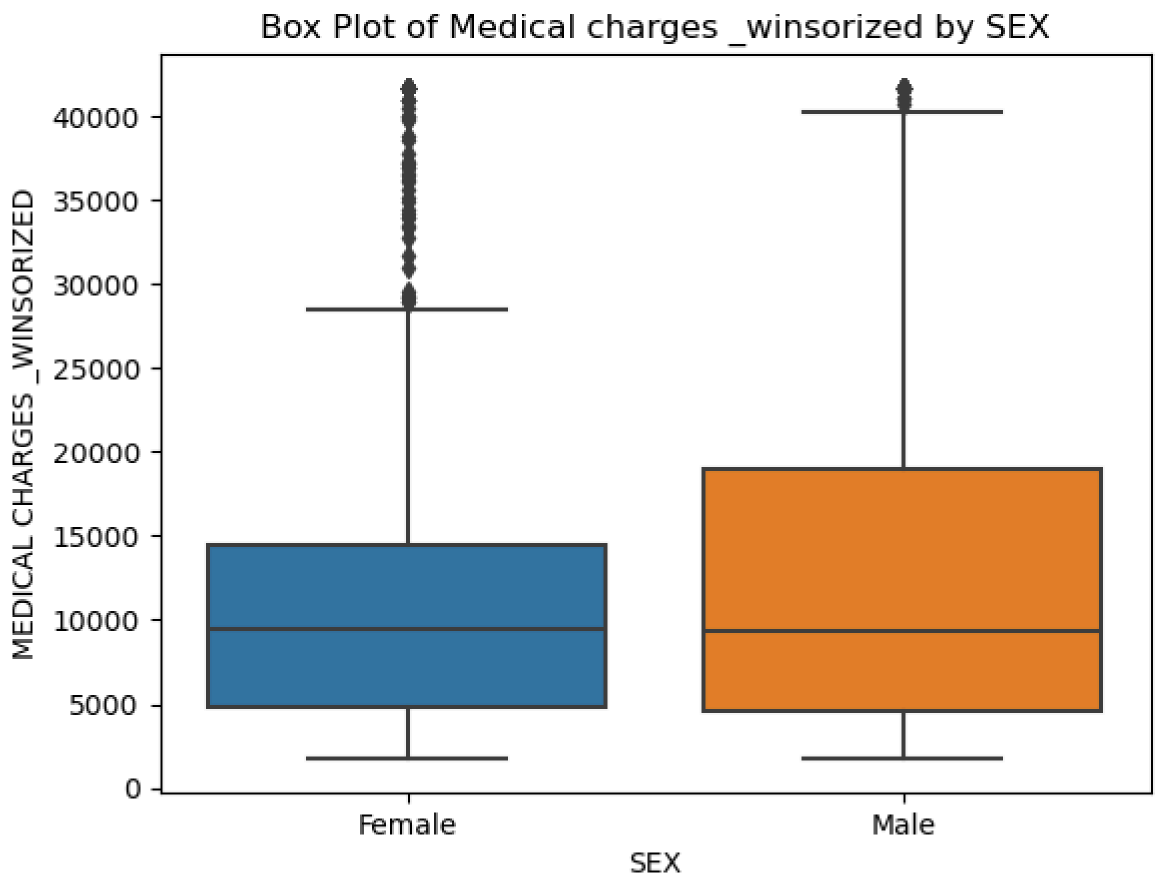
In [19]:
```
from scipy.stats.mstats import winsorize
```

In [20]:
```
df['medical charges_winsorized'] = winsorize(df['medical charges'], limits=(0.05,
```

In [21]:
```
sns.boxplot(x=df['sex'], y=df['medical charges_winsorized'])

plt.title('Box Plot of Medical charges _winsorized by SEX')
plt.xlabel('SEX')
plt.ylabel('MEDICAL CHARGES _WINSORIZED')

plt.show()
```



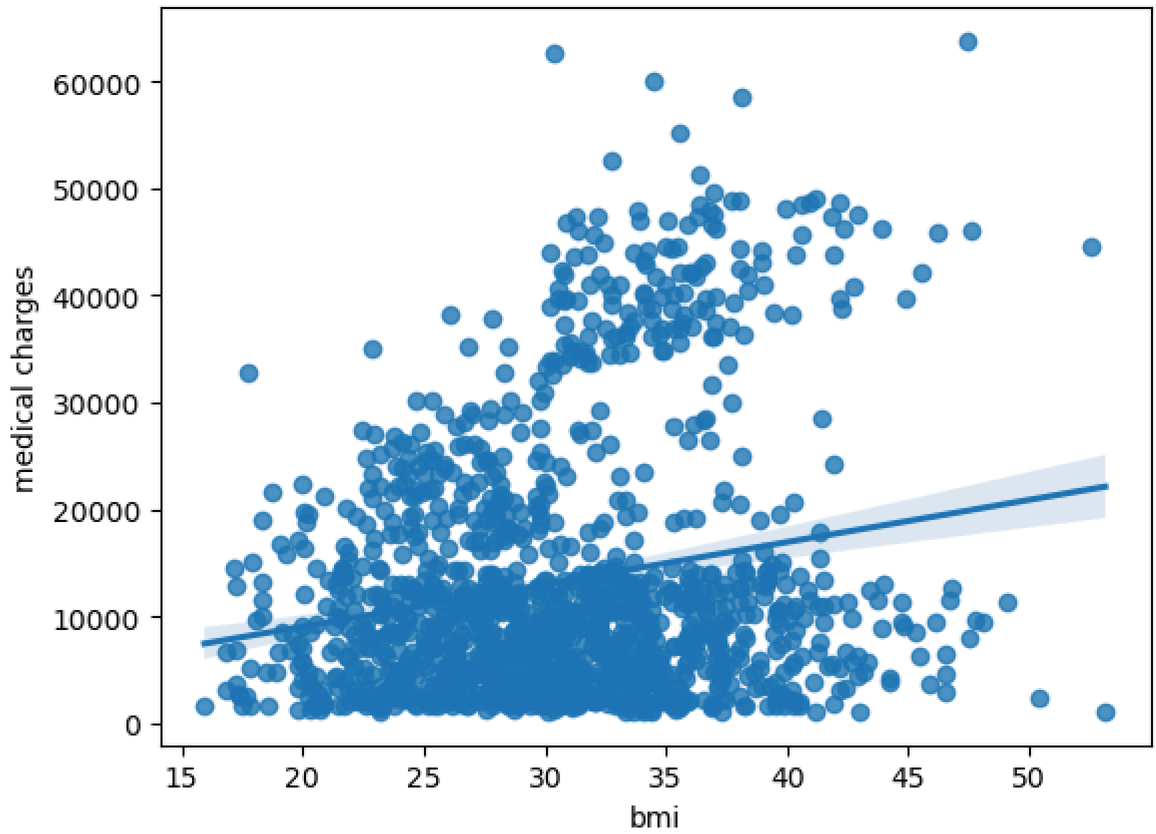
as we can see our outlier have reduce and our data is in a good shape

let's check the linear correlation of our data set using visualization

For age, bmi, children, smoker_encoded to medical charges_winsorized

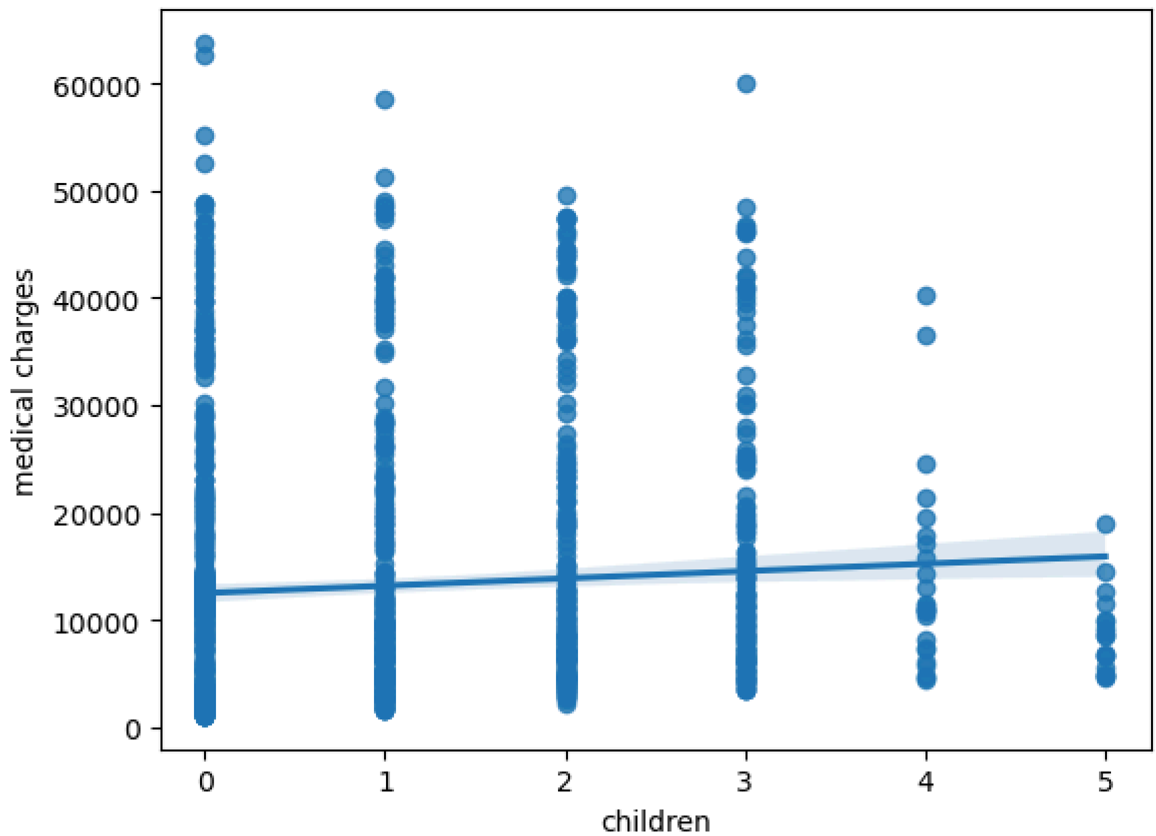
In [22]:
```
sns.regplot(x="bmi", y="medical charges", data=df)
```

Out[22]:
```
<Axes: xlabel='bmi', ylabel='medical charges'>
```



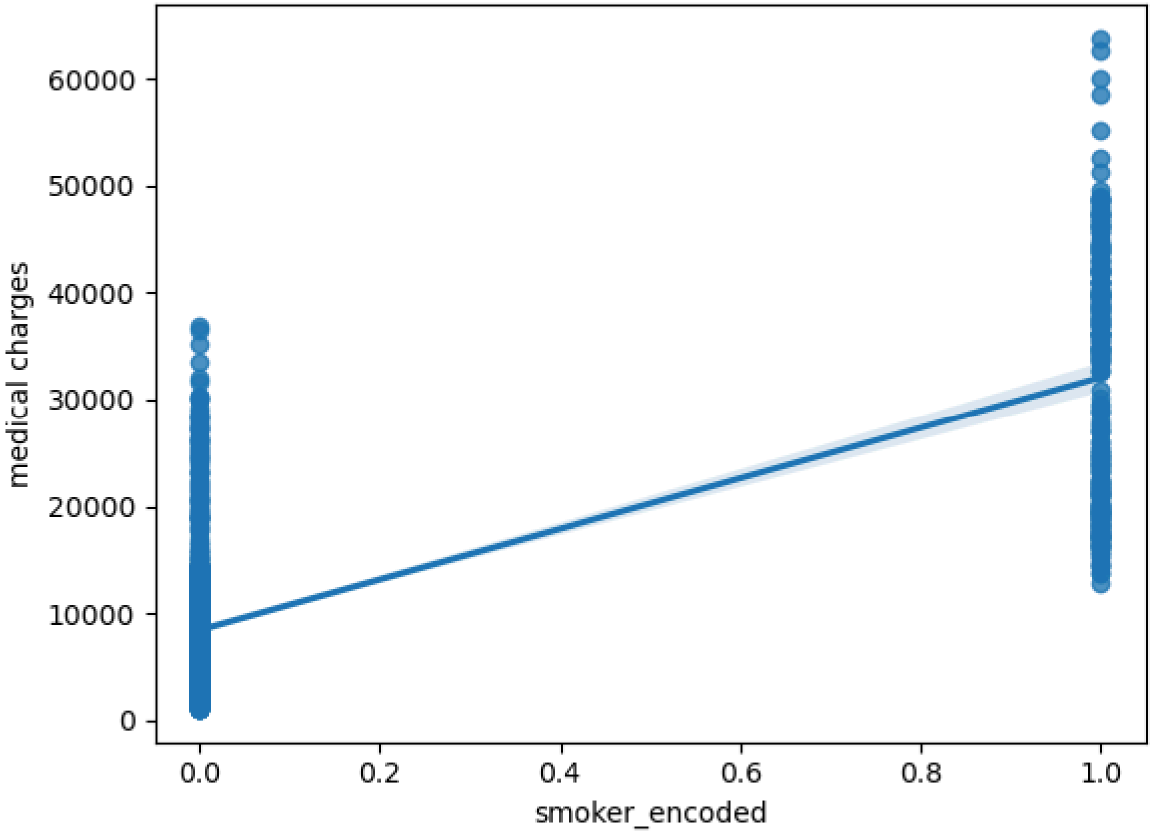
In [23]:
```
sns.regplot(x="children", y="medical charges", data=df)
```

Out[23]:
```
<Axes: xlabel='children', ylabel='medical charges'>
```



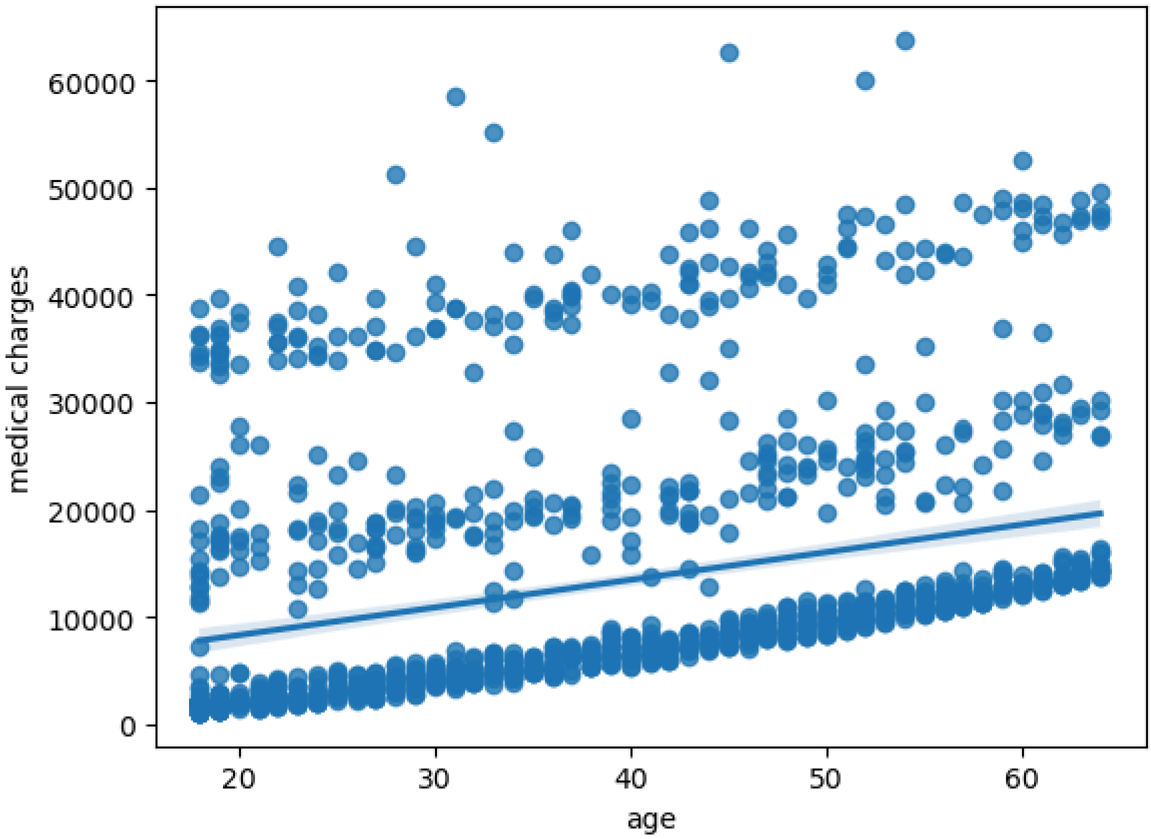
In [24]:
```
sns.regplot(x="smoker_encoded", y="medical charges", data=df)
```

Out[24]:
```
<Axes: xlabel='smoker_encoded', ylabel='medical charges'>
```



In [25]:
```
sns.regplot(x="age", y="medical charges", data=df)
```

Out[25]:
```
<Axes: xlabel='age', ylabel='medical charges'>
```



Let's check the correlation using pearsonr

In [26]:
```
import scipy.stats
from scipy.stats import pearsonr
```

In [27]:
```
pearson_coef, p_value = pearsonr(df["bmi"], df["medical charges_winsorized"])
print("the pearson correlation coefficient is", pearson_coef, "with p_value of p='
```

```
the pearson correlation coefficient is 0.18687299633023485 with p_value of p= 5.580
832278700515e-12
```

The independent variable(BMI) has a weak positive linear relationship and a statistically significant effect on the dependent variable (medical charges). the magnitude of the effect (0.187) is relatively small(weak).

In [28]:
```
pearson_coef, p_value = pearsonr(df["children"], df["medical charges_winsorized"])
print("the pearson correlation coefficient is", pearson_coef, "with p_value of p='
```

```
the pearson correlation coefficient is 0.07108757442313804 with p_value of p= 0.009
29116180280871
```

The independent variable(children) has a very weak positive linear relationship and a statistically significant effect on the dependent variable (medical charges). the magnitude of the effect (0.07) which indicate a very weak relationship.

In [29]:
```
pearson_coef, p_value = pearsonr(df["smoker_encoded"], df["medical charges_winsor
print("the pearson correlation coefficient is", pearson_coef, "with p_value of p='
```

```
the pearson correlation coefficient is 0.7929162561913092 with p_value of p= 1.0072
8723433396547e-289
```

The independent variable(smoker_encoded) has a strong positive linear relationship and a statistically significant effect on the dependent variable (medical charges). the magnitude of the effect is (0.79). which is relatively large.

In [30]:
```
pearson_coef, p_value = pearsonr(df["age"], df["medical charges_winsorized"])
print("the pearson correlation coefficient is", pearson_coef, "with p_value of p='
```

```
the pearson correlation coefficient is 0.29774275703151154 with p_value of p= 8.537
14982863166e-29
```

The independent variable(age) has a weak positive linear relationship and a statistically significant effect on the dependent variable (medical charges). the magnitude of the effect is (0.29). which is relatively weak.

- let's group and sum our data set using region, smoker, by smoker_encoded to see which region has more smoker

In [31]:
```
df.groupby(["region", "smoker"], as_index=False)["smoker_encoded"].sum()
```

Out[31]:

	region	smoker	smoker_encoded
0	Northeast	No	0
1	Northeast	Yes	67
2	Northwest	No	0
3	Northwest	Yes	58
4	Southeast	No	0
5	Southeast	Yes	91
6	Southwest	No	0
7	Southwest	Yes	58

southeast has the highest smokers

In [32]:
```
df.groupby(["region"], as_index=False)["medical charges"].sum()
```

Out[32]:

	region	medical charges
0	Northeast	4.343669e+06
1	Northwest	4.035712e+06
2	Southeast	5.363690e+06
3	Southwest	4.012755e+06

we can see that the region with highers number of smokers is southeast with the total number of 91 individuals that smokers

- let's group and sum our data set using region, smoker, by medical charges to see which region pay more medical charges

In [33]:
```
df.groupby(["region", "smoker"], as_index=False)["medical charges"].sum()
```

Out[33]:

	region	smoker	medical charges
0	Northeast	No	2.355542e+06
1	Northeast	Yes	1.988127e+06
2	Northwest	No	2.284576e+06
3	Northwest	Yes	1.751136e+06
4	Southeast	No	2.192795e+06
5	Southeast	Yes	3.170895e+06
6	Southwest	No	2.141149e+06
7	Southwest	Yes	1.871606e+06

Here we can see that the region with highest medical charges is southeast region with 3,170,895.00 and it appears to be region with highest individual that smoke.

FINDING

After analyzing the linear relationships, we concluded that age, children, and BMI have a limited or very small influence on medical charges, even though their effects are statistically significant. In contrast, the variable smoker shows a much stronger influence on medical charges.

This suggests that while age, children, and BMI do have some statistical significance in relation to medical charges, their actual impact on the prices is minimal. Conversely, the smoking status has a more substantial effect on medical charges, indicating that it is a more critical factor in determining healthcare costs. and it is also the major factor that causes the increased in medical charges

lets developt our model

Here we are using multiple linear regression model to analyse our model we will focus on four variable for developing our model. medical charges winsirozed, age, bmi, and smoker status

Create the linear regression object:

In [34]:
```
lm = LinearRegression()
lm
```

Out[34]:
```
▾ LinearRegression
LinearRegression()
```

train my model

In [35]:
```
Z = df[["age", "bmi",  "smoker_encoded"]]
```

In [36]:
```
lm.fit(Z, df["medical charges_winsorized"])
```

Out[36]:
```
▾ LinearRegression
LinearRegression()
```

In [37]:
```
lm.intercept_
```

Out[37]:
```
-9834.137830725183
```

In [38]:
```
lm.coef_
```

Out[38]:
```
array([  244.5113938 ,   282.27750166, 22594.84046598])
```

medical charges = -9834.137830725183 + 244.5113938*(age)* + *282.27750166*(bmi) + 22594.84046598*(smoker status)

- Let's Find the R-square

In [39]:
```
lm.fit(Z, df['medical charges_winsorized'])
print('The R-square is: ', lm.score(Z, df['medical charges_winsorized']))
```

```
The R-square is:  0.7521507710816308
```

this mean 75% of the variability in the outcome variable (medical charges) is explained by the predictor(age, bmi & smoker status) which was included in the model. Although there is still almost 25% of the variability that is not explained by the model. This remaining variability could be due to other factors not included in the model. this indicate that age, bmi and smoker status have a significant contribution to the outcome(medical charges)

lets evaluate our model

- using Training and Testing method

we will train or teach our machine learning model with % of our data and test it with % of our data set to evaluate the performance of the model

In [40]:
```
x = df[["age", "bmi", "smoker_encoded"]]
y = df["medical charges_winsorized"]
```

In [41]:
```
x_data=df.drop('medical charges_winsorized',axis=1)
```

In [42]:
```
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.3, random_st

print("number of test samples :", x_test.shape[0])
print("number of training samples:",x_train.shape[0])
```

```
number of test samples : 402
number of training samples: 936
```

In [43]:
```
model=LinearRegression()
```

In [44]:
```
model.fit(x_train, y_train)
```

Out[44]:
```
▾ LinearRegression
LinearRegression()
```

In [45]:
```
y_train_pred = model.predict(x_train)
y_test_pred = model.predict(x_test)
```

In [46]:
```
model.score(x_train, y_train)
```

Out[46]: 0.7555544031806487

In [47]:
```
model.score(x_test, y_test)
```

Out[47]: 0.7426441931512848

In [48]:
```
mse_train = mean_squared_error(y_train, y_train_pred)
mse_test = mean_squared_error(y_test, y_test_pred)
```

In [49]:
```
print("MSE on Training Set:", mse_train)
print("MSE on Testing Set:", mse_test)
```

```
MSE on Training Set: 32409513.79155388
MSE on Testing Set: 31855831.843142353
```

The Mean Squared Error for the training and testing se is very close. This close MSE between training and testing data indicates that the model is not overfitting, as it performs almost as well on unseen data as it does on the training data.

Results & Interpretation

The model currently focuses on age, BMI, and smoker status as predictor variables, but additional factors like gender, region, and the number of beneficiary children could provide a more comprehensive analysis. For example, healthcare costs often vary by gender due to differences in health risks and gender-specific medical needs. Regional variations can also impact medical costs because of differences in healthcare systems, access to services, or demographic characteristics in different locations. Furthermore, the number of children covered under a policy can influence medical costs, as more beneficiaries might increase expenses.

In the current model, we use age, BMI, and smoker status as independent variables to predict the dependent variable, which is medical charges. Based on our analysis, the model explains approximately 75% of the variation in medical charges. This indicates that while the selected variables have a strong relationship with medical costs, some portion of the variance remains unexplained, and that could improve the model's accuracy.

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